43.

Element	%	Relative no. of atoms	Simple ratio
Na	14.31	$\frac{14.31}{23} = 0.62$	$\frac{0.62}{0.31} = 2$
S	9.97	$\frac{9.97}{32} = 0.31$	$\frac{0.31}{0.31} = 1$
H	6.22	$\frac{6.22}{1} = 6.22$	$\frac{6.22}{0.31} = 20$
O	69.5	$\frac{69.5}{16} = 4.34$	$\frac{4.34}{0.31} = 14$

$\therefore$ Empirical formula = $Na_2\ S\ H_{20}\ O_{14}$

$$n = \frac{\text{molar mass}}{\text{calculated empirical formula mass}} = \frac{322}{322} = 1$$

$$\begin{aligned} & Na_2\ S\ H_{20}\ O_{14} \\ &= (2\times 23) + (1\times 32) + (20\times 1) + 14\,(16) \\ &= 46 + 32 + 20 + 224 \\ &= 322 \end{aligned}$$

Molecular formula = $Na_2\ S\ H_{20}\ O_{14}$

Since all the hydrogen in the compound present as water

$\therefore$ Molecular formula is $Na_2\ SO_4\ .\ 10H_2O$

44. (i)

$$K_2\overset{+6}{Cr_2}O_7 + K\overset{-1}{I} + H_2SO_4 \rightarrow K_2SO_4 + \overset{+3}{Cr_2}(SO_4)_3 + \overset{0}{I_2} + H_2O$$

$2 \times 3e^-$ $\quad$ $1e^-$

$$\mathbf{K_2Cr_2O_7} + \mathbf{6KI} + H_2SO_4 \rightarrow K_2SO_4 + Cr_2\,(SO_4)_3 + I_2 + H_2O$$

$$K_2Cr_2O_7 + 6KI + H_2SO_4 \rightarrow K_2SO_4 + \mathbf{Cr_2(SO_4)_3} + \mathbf{3I_2} + H_2O$$

$$K_2Cr_2O_7 + 6KI + \mathbf{7H_2SO_4} \rightarrow \mathbf{4K_2SO_4} + Cr_2\,(SO_4)_3 + 3I_2 + \mathbf{7H_2O}$$

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